

CHEONGJU, Korea, Republic of

WMO: 471310

Lat: 36.6392N

Lon: 127.4406E

Elev: 192

StdP: 14.59

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.0	16.4	-2.7	4.8	21.0	0.6	5.7	22.6	16.2	28.6	14.2	28.7	3.2	290	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.3	91.7	76.1	89.4	74.9	87.1	73.6	78.6	86.2	77.5	84.9	76.5	83.8	6.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.6	139.6	82.1	75.5	134.4	81.5	74.4	129.4	81.1	42.3	86.3	41.2	85.1	40.2	83.8	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.2	9.6	8.4	94.2	4.4	2.5	5.2	96.0	2.7	97.5	0.2	98.9	-3.0	100.6		
			6.6	79.6	3.9	2.0	3.8	81.0	1.5	82.2	-0.7	83.3	-3.5	84.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	29.5	34.1	43.9	55.8	66.0	73.6	79.0	79.8	70.9	59.4	46.1	33.5
	DBStd	18.47	6.87	6.86	6.76	6.53	4.79	3.69	4.16	4.57	4.93	6.01	7.35	7.14
	HDD50	1993	635	445	211	21	0	0	0	0	0	8	160	513
	HDD65	4692	1100	864	654	283	46	1	0	0	9	189	568	978
	CDD50	4208	0	1	22	194	494	707	900	923	626	298	43	0
	CDD65	1433	0	0	0	7	76	258	435	458	185	14	0	0
	CDH74	12008	0	0	0	77	565	1739	3993	4523	1062	49	0	0
	CDH80	3742	0	0	0	6	86	424	1337	1686	202	1	0	0
	WSAvg	3.7	3.4	3.7	4.4	4.5	4.3	3.9	4.0	3.9	3.5	3.1	3.1	3.1
	PrecAvg	47.6	1.2	1.3	2.0	3.2	3.4	5.4	10.8	9.8	5.6	1.8	1.9	1.1
Precipitation	PrecMax	66.0	3.9	3.9	6.7	7.4	8.2	14.3	21.1	22.5	13.7	6.2	5.0	2.7
	PrecMin	33.5	0.0	0.0	0.1	0.6	0.7	0.2	5.0	3.1	0.2	0.1	0.4	0.1
	PrecStd	9.1	1.0	1.0	1.3	2.0	1.8	3.8	4.0	5.1	3.5	1.5	1.2	0.6
	PrecStd	9.1	1.0	1.0	1.3	2.0	1.8	3.8	4.0	5.1	3.5	1.5	1.2	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	51.1	58.6	68.9	80.1	85.8	88.9	94.8	95.2	87.7	78.4	69.2	55.2	
	0.4% MCWB	44.2	50.3	52.3	59.6	62.7	67.5	77.6	76.8	73.0	63.9	57.3	46.9	
	2% DB	45.5	53.0	64.0	76.1	82.2	86.6	91.4	92.5	84.6	75.1	64.4	50.5	
	2% MCWB	38.8	42.9	49.5	57.4	61.8	67.4	76.4	76.3	70.4	61.5	54.8	43.3	
	5% DB	42.3	49.2	60.1	72.4	79.8	84.5	89.1	90.3	82.3	72.4	61.1	47.2	
	5% MCWB	36.4	40.7	47.2	55.1	60.9	66.9	75.5	75.8	69.1	60.3	52.3	41.0	
	10% DB	39.7	45.4	56.4	68.8	77.2	82.4	86.8	88.0	80.0	69.8	57.9	44.3	
	10% MCWB	34.5	37.9	45.5	53.6	59.8	66.7	74.8	75.2	67.9	59.0	50.1	39.1	
	0.4% WB	44.1	52.0	56.0	65.4	69.9	75.2	79.7	80.2	77.1	68.8	61.0	48.4	
	0.4% MCDB	48.4	56.4	62.6	73.7	75.3	80.0	88.1	88.7	83.5	73.3	66.0	53.7	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	40.3	44.5	52.6	61.1	66.1	72.8	78.3	79.0	74.4	65.3	56.6	44.5	
	2% MCDB	44.5	50.4	60.9	72.4	75.8	78.6	85.9	86.9	80.1	71.1	62.7	48.8	
	5% WB	37.4	41.2	49.1	58.2	64.3	71.4	77.4	78.0	72.6	62.7	53.7	41.9	
	5% MCDB	41.0	48.1	57.4	67.7	74.0	78.5	84.9	85.5	78.6	68.9	59.1	46.5	
	10% WB	34.9	38.6	46.4	55.5	62.6	70.0	76.5	76.9	70.6	60.7	51.2	39.4	
	10% MCDB	39.1	44.5	55.0	65.5	72.6	77.8	83.8	84.2	77.0	67.5	57.0	44.2	
Mean Daily Temperature Range	MDBR	14.1	16.1	18.2	20.3	19.0	15.4	11.8	12.3	15.0	17.8	16.3	14.3	
	5% DB MCDBR	16.3	21.2	24.3	26.4	23.4	19.3	15.0	15.0	17.1	20.4	20.4	18.0	
	5% DB MCWBR	11.2	13.1	12.9	11.2	8.4	6.2	4.9	4.6	6.0	8.7	11.6	12.4	
	5% WB MCDBR	14.4	18.7	20.1	20.7	17.1	13.4	12.3	12.5	12.7	15.1	16.4	16.6	
	5% WB MCWBR	11.1	13.2	12.6	11.0	8.1	6.0	4.7	4.8	5.9	8.5	11.3	12.5	
	5% WB MCWBR	11.1	13.2	12.6	11.0	8.1	6.0	4.7	4.8	5.9	8.5	11.3	12.5	
Clear-Sky Solar Irradiance	taub	0.436	0.482	0.567	0.598	0.610	0.678	0.631	0.571	0.496	0.480	0.464	0.432	
	taud	1.944	1.870	1.715	1.696	1.712	1.636	1.793	1.936	2.077	1.993	1.972	1.975	
	Ebn at Noon	224	230	222	225	226	210	219	230	240	228	213	215	
	Edh at Noon	45	55	70	75	75	81	69	58	48	48	43	41	
All-Sky Solar Radiation	RadAvg	789	1027	1347	1561	1760	1611	1316	1366	1270	1130	793	702	
	RadStd	43	83	137	126	155	137	181	197	141	89	91	50	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.90	N/A	-1.80	N/A	N/A	N/A	N/A	N/A	+263	+149	
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+81	

Nomenclature: See separate page